

Project Title:

Customer Feedback Sentiment Analysis and Automated Response Generation System

Problem Statement

In most organizations, customer feedback and complaint management are handled manually. When customers share negative experiences such as poor product quality, delayed services, or unhelpful support, their concerns often go unnoticed or receive delayed responses. This not only leads to customer dissatisfaction but also negatively affects brand reputation and retention rates.

Despite having large volumes of feedback data, companies struggle to efficiently identify dissatisfied customers and respond to them in a timely and personalized manner.

Objective

The primary objective of this project is to design and develop an intelligent system that automatically analyzes customer feedback, identifies dissatisfied customers using sentiment analysis, and generates personalized acknowledgment emails to enhance customer engagement and retention.

The system aims to assist companies in improving their customer service efficiency, reducing response time, and ensuring that every customer receives a professional and empathetic response.

Proposed Solution

The proposed system will:

1. **Collect and Store Feedback:**

Gather customer feedback data in a structured database (Excel or any suitable format) containing customer IDs, comments, and email addresses.

2. **Perform Sentiment Analysis:**

Utilize the **Gemini API** (or similar NLP model) to classify each feedback comment into five sentiment categories:

Very Satisfied, Satisfied, Neutral, Dissatisfied, and Very Dissatisfied.

3. **Generate Personalized Email Responses:**

Automatically draft customized email replies acknowledging customer feedback.

- Positive feedback will receive appreciation messages.
- Dissatisfied customers will receive empathetic emails along with optional promotional coupons or discounts to rebuild trust.

4. **Provide Employee Dashboard:**

Create a simple **user interface (UI)** for company representatives to review negative feedback, view or edit generated emails, and send them directly to customers.

Purpose and Significance

This project bridges the gap between **customer sentiment detection** and **automated communication**. By integrating NLP-driven sentiment analysis with automated response generation:

- Companies can **quickly identify at-risk customers** and prevent potential churn.
 - Employees can **save time** by focusing only on high-priority feedback.
 - Customers receive **faster and more personalized responses**, improving their overall experience.
 - The system can later be scaled to real-time operations, providing **data-driven insights** into customer satisfaction trends.
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Expected Outcome

A functional prototype that:

- Accurately classifies customer feedback sentiments using AI.
- Automatically drafts relevant email responses.
- Provides a clean, interactive UI for employees to manage customer communication efficiently.